

Frequent mistakes on Quizzes:

Quiz 1:

(pretty same mistakes on both Ro and Eng tracks)

1. Evaluating complexity – problems for evaluating time on mostly exponential but also some for linear, like: “For an exponential/linear running time algorithm solving a given problem of size greater than ...”:
2. QuickSort – problems asking questions on different aspects on the strategy, especially when particular cases (best/worst for given implementations) are asked.

Quiz 2:

(Ro test)

1. Evaluating complexity (similar to Quiz 1) + Optimality sorting questions + lower bound meaning
2. Balanced trees questions – like “This is the preorder of a BST Which of the following statements is true?” and different types of balances provided.

(Eng test)

1. Balanced trees questions – like “This is the preorder of a BST Which of the following statements is true?” and different types of balances provided.
2. Balanced trees – different kind of balances a tree has.

Quiz 3:

(Ro test)

1. mistakes mainly on questions regarding: RB trees or OS trees or Disjoint Sets.
2. mistakes mainly on questions on: (i) what type of balances (among PBT, AVL, RB) a GIVEN tree has; (ii) WHY a GIVEN tree is NOT of a GIVEN balance type, (iii) modalities of fixing the issue to turn a tree into a GIVEN balance structure

(Eng test)

1. mistakes mainly on questions regarding: RB trees or OS trees.
2. most frequent on: (i) potential type of an added edge on G, (ii) different possible edge classifications on different dfs traversals of G, (iii) Edge type transformation in dfs (like in “In the following graph the dfs which considers the lexicographical order of vertices would classify (a, b) as **tree** edge. Which order should we consider so that it becomes a”) (iv) number of dfs trees (like in “Given the graph G... the minimum number of **dfs trees** obtained (that is, regardless how vertices and edges are ordered) is: ...”

Quiz 4:

(Ro test)

1. mistakes mainly on questions regarding the review on Data Structures: RB trees or OS trees or Disjoint Sets.
2. most frequent on: (i) Edge type transformation in dfs (like in “In the following graph the dfs which considers the lexicographical order of vertices would classify (a, b) as **tree** edge. Which order should we consider so that it becomes a”) (ii) number of dfs trees (like in “Given the graph G... the minimum number of **dfs trees** obtained (that is, regardless how vertices and edges are ordered) is:” (iii) potential type of an added edge on G,

(Eng test)

1. SSC: Qs on: (i) number of SCC in a directed graph; (ii) number of nodes in the largest SSC in a directed graph
2. DFS: Qs on: (i) which node turns first black on a dfs;

Quiz5:

(Ro test)

1. SSC Qs on: (i) number of SCC in a directed graph; (ii) number of nodes in the largest SSC in a directed graph
2. DFS: Qs on: (i) which node turns first black on a dfs;

The average of the quizzes grade was 4.98, with a minimum of 0 and a maximum of 9.58 (we took into account the students that participated in at least one quiz). You can see the grades distribution in the figure below.

